

Alex Weng

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EDUCATION

CORNELL UNIVERSITY, College of Engineering

Ithaca, NY

Major: Mechanical Engineering | Minor: Aerospace Engineering | GPA: 3.73

August 2023 - May 2027

Relevant Courses: Statics | Dynamics | System Dynamics | Structural Dynamics and Vibrations Analysis | Heat Transfer | Intro to Mechanical Design | Intro to Aeronautics | FEA for Mechanical and Aerospace Design | Mechanics of Materials | Mechatronics | Introductory Fluid Mechanics | Thermodynamics | Uncertainty Analysis in Engineering

Activities: Cornell Unmanned Air Systems (Full Team Safety Lead) | MAE 3260 (System Dynamics) TA | Symbiotic Engineering and Analysis Lab (Research) | Cornell Cycling Club (3-year officer) | Campus Tour Guide

EXPERIENCE

The Boeing Company, Everett, WA

May 2026 - August 2026

Loads and Dynamics Engineering Intern - 737 MAX Dynamic Flight Loads and Flutter

- Established CG acceleration thresholds to waive unnecessary inspections after pilot-reported severe turbulence
- Simulated aircraft response to wind gusts using Easy5 and internal Boeing tools on Linux HPC
- Conducted a survey of 10,000+ gust conditions utilizing a B737-MAX dynamic loads model
- Utilized Matlab to plot airplane CG response vs turbulence frequency and vs discrete gusts at various gust gradients
- Conducted parameter variation analyses to characterize growth rates of B737 horizontal stabilizer flutter modes
- Compared NG and MAX mode shapes and frequencies using NASTRAN models, visualized flutter modes on Patran

Cornell Unmanned Air Systems Project Team, Ithaca, NY

February 2025 - Present

Structures & Payloads Engineer

- Designed and fabricated GoPro gimbal (100° roll/pitch) achieving 70% weight and 55% torque reduction vs. prior model
- Modeled full assembly in SolidWorks; selected servos, bearings, and dowel pins for strength and manufacturability
- Integrated gimbal hardware with aircraft avionics; collaborated with electrical and imaging teams on interface design
- Manufactured and tested 4+ gimbal prototypes using 3D-printed and ordered parts, resolving issues related to GoPro fit, bearing motion, part connections, center of gravity of rotating parts, and space constraints within the aircraft
- Cut bay doors in carbon composite aircraft fuselage using a Dremel; designed, installed, and tested custom bay door

Sibley School of Mechanical and Aerospace Engineering, Ithaca, NY

August 2026 - Present

Lab Instructor for MAE 3620 - System Dynamics

- Lead 20-student system dynamics labs utilizing LabVIEW VIs, oscilloscopes, MATLAB, shakers, motors, and rotor rigs
- Teach students to characterize and balance high speed rotary systems and design functional feedback control systems
- Guide frequency-response and modal testing to identify resonance and vibration modes of beams and rotors

Symbiotic Engineering and Analysis Lab at Cornell University, Ithaca, NY

July 2024 - September 2025

Undergraduate Research Assistant

- Designed and validated mounts for a 94-lb pressure vessel and 25-lb water pump using Fusion 360 and FEA
- Assembled full hydraulic testing system (6 valves, 9 sensors, 40+ fittings) for high-pressure water treatment experiments
- Brainstormed schematics of a WEC-driven carbon sequestration system that pumps 1 GtCO₂/year to depth of ~3000m
- Made design decisions based on suitability of different sequestration mediums and methods with WEC power
- Assessed optimization methods for RM6 WEC architecture through literature review, informing design tradeoffs

Cornell University Systems Engineering, Ithaca, NY

June 2025 - August 2025

Summer Research Assistant

- Simulated hybrid wind-wave energy converter dynamics using MATLAB, Wec-Sim, and ANSYS AQWA
- Modeled hydrodynamic bodies on CAD and evaluated stability and power capture efficiency of various architectures
- Built Simulink models of each configuration; wrote scripts in MATLAB to analyze their performance

SKILLS & INTERESTS

Software: SolidWorks, Fusion 360, AutoCAD, MATLAB, ANSYS (FEA/AQWA), Simulink, Easy5, Linux OS, Patran, MS Office Suite

Core Skills: CAD Modeling, Mechanical Design, Rapid Prototyping, Finite Element Analysis (FEA), Modal Analysis

Interests: Mechanical Design, Structural Dynamics, Vibrations, Sustainability, Entrepreneurship, Cycling, Soccer